

BS-CH101 / BS-CH201 — Chemistry-I

Semester I • 3L:1T:0P • 4 Credits

Module 5 Exam Notes: Periodic Properties

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. Periodic Properties

Periodic properties are atomic and chemical properties that vary systematically with atomic number due to effective nuclear charge, shielding, and electronic configuration.

1.1 Effective Nuclear Charge and Shielding

Outer electrons do not experience the full nuclear charge because inner electrons shield them. Effective nuclear charge is:

Effective Nuclear Charge

$$Z_{\text{eff}} = Z - \sigma$$

where Z is nuclear charge and σ is shielding constant.

1.2 Penetration and Orbital Energy

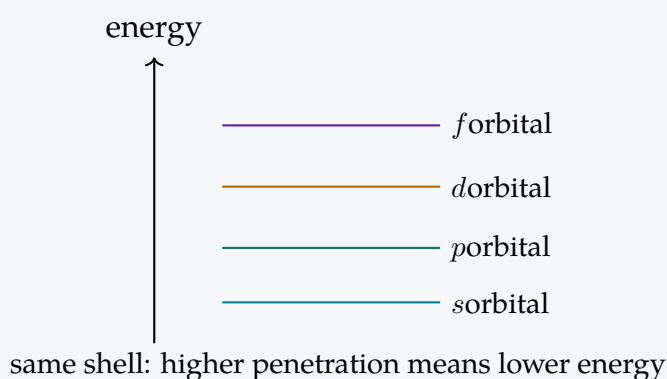
For the same principal shell, penetration order is:

$$s > p > d > f$$

More penetrating orbitals experience higher Z_{eff} and lower energy. Therefore in multi-electron atoms:

$$E(ns) < E(np) < E(nd) < E(nf)$$

Penetration and Energy Order



1.3 Atomic Size, Ionization Energy, Electron Affinity and Electronegativity

Across a period, Z_{eff} increases, so atomic size decreases and ionization energy generally increases. Down a group, new shells are added, so atomic size increases and ionization energy decreases.

Property	Across period	Down group
Atomic radius	Decreases	Increases
Ionization energy	Generally increases	Decreases
Electron affinity	Becomes more negative generally	Less negative generally
Electronegativity	Increases	Decreases
Metallic character	Decreases	Increases

1.4 Polarizability, Oxidation States and Coordination

Polarizability increases with size and diffuse electron cloud. Oxidation state depends on valence electron configuration. Coordination number and geometry depend on metal ion size, ligand size, and electron count.

Coordination number	Common geometry	Example type
2	Linear	$[\text{Ag}(\text{NH}_3)_2]^+$
4	Tetrahedral or square planar	$[\text{Ni}(\text{CN})_4]^{2-}$
6	Octahedral	$[\text{Fe}(\text{CN})_6]^{3-}$

1.5 Hard and Soft Acids and Bases

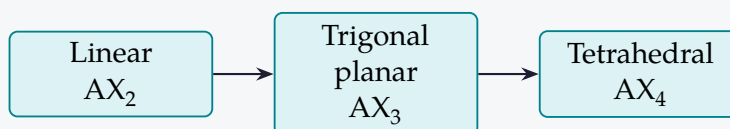
HSAB principle says hard acids prefer hard bases and soft acids prefer soft bases.

Hard species are small, highly charged, and weakly polarizable. **Soft species** are larger, lower charged, and highly polarizable.

1.6 Molecular Geometries

VSEPR theory predicts geometry from electron-pair repulsion around the central atom.

Common VSEPR Shapes



Exam Points

- Explain periodic trends using Z_{eff} , shielding, and shell number.
- For orbital energy, mention penetration order $s > p > d > f$.
- HSAB questions need definition plus examples.
- Geometry questions should mention electron pairs before shape.

2. Detailed Periodic Trend Analysis

2.1 Shielding Order and Slater-Type Reasoning

Shielding is not equal for all orbitals. Inner-shell electrons shield strongly, while electrons in the same shell shield only partially. Poor shielding by *d* and *f* electrons explains many irregular periodic trends.

Electron type	Shielding ability	Consequence
<i>s</i> electron	Strong penetration and good shielding	Feels high Z_{eff}
<i>p</i> electron	Moderate shielding	Higher energy than same-shell <i>s</i>
<i>d</i> electron	Poor shielding	Causes transition-series contraction
<i>f</i> electron	Very poor shielding	Causes lanthanide contraction

2.2 Successive Ionization Energies

Successive ionization energies always increase because after removal of one electron, the remaining electrons are held more strongly by the positive ion.

$$\text{IE}_1 < \text{IE}_2 < \text{IE}_3 < \dots$$

A sudden jump in ionization energy indicates removal of an electron from an inner shell.

Solved Example: Identifying Valence Electrons

If successive ionization energies show the pattern:

$$580, \quad 1820, \quad 2750, \quad 11600 \text{ kJ mol}^{-1}$$

the large jump occurs after the third electron. Therefore the atom has **three valence electrons**.

2.3 Diagonal Relationship

Some second-period elements resemble third-period elements diagonally placed in the periodic table. Examples include Li–Mg, Be–Al, and B–Si. This occurs because size and charge density become comparable.

Pair	Common similarities	Reason
Li and Mg	Form nitrides; carbonates decompose on heating	Similar charge density
Be and Al	Amphoteric oxides and covalent halides	Small size and high polarizing power
B and Si	Covalent compounds and acidic oxides	Similar electronegativity trend

2.4 Oxidation States and Inert Pair Effect

For heavier *p*-block elements, the outer ns^2 electron pair becomes less available for bonding. This is called inert pair effect. It stabilizes lower oxidation states.

Inert pair effect: Reluctance of the outermost ns^2 electron pair of heavier elements to participate in bonding.

Group	Higher oxidation state	Lower oxidation state stabilized down group
Group 13	+3	+1 in Tl
Group 14	+4	+2 in Pb
Group 15	+5	+3 in Bi

2.5 HSAB Applications

HSAB theory predicts stability of complexes and salts. Hard-hard and soft-soft combinations are more stable than hard-soft combinations.

Acid / base type	Examples	Preferred partner
Hard acids	H^+ , Li^+ , Al^{3+} , Fe^{3+}	Hard bases
Soft acids	Ag^+ , Hg^{2+} , Pt^{2+}	Soft bases
Hard bases	F^- , OH^- , H_2O , NH_3	Hard acids
Soft bases	I^- , S^{2-} , CO , PR_3	Soft acids

Solved Example: HSAB Prediction

Ag^+ is a soft acid. Between F^- and I^- , iodide is softer. Therefore AgI is more stable and less soluble than expected from simple charge arguments.

2.6 VSEPR Shapes and Lone Pair Effect

Lone pairs repel more strongly than bond pairs. The repulsion order is:

$$\text{lone pair-lone pair} > \text{lone pair-bond pair} > \text{bond pair-bond pair}$$

Molecule type	Shape	Example
AX_2	Linear	$BeCl_2$, CO_2
AX_3	Trigonal planar	BF_3
AX_3E	Trigonal pyramidal	NH_3
AX_2E_2	Bent	H_2O
AX_4	Tetrahedral	CH_4

Periodic Properties: 10-Mark Answer Framework

1. Define the property.
2. State the trend across period and down group.
3. Explain trend using Z_{eff} , shielding, and shell size.

4. Mention exceptions if relevant.
5. Add one example or application such as HSAB or VSEPR shape.

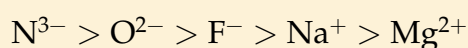
2.7 Atomic and Ionic Radius: Detailed Trend

Atomic radius decreases across a period because effective nuclear charge increases while electrons enter the same shell. Down a group, radius increases because a new shell is added.

Species compared	Trend	Reason
Cations vs atoms	Cations are smaller	Electron loss reduces repulsion and may remove outer shell
Anions vs atoms	Anions are larger	Extra electron increases electron-electron repulsion
Isoelectronic species	Radius decreases with increasing nuclear charge	Same electrons pulled by larger Z

Solved Example: Isoelectronic Radius Order

N^{3-} , O^{2-} , F^- , Na^+ and Mg^{2+} are isoelectronic with 10 electrons.
Higher nuclear charge pulls the same electron cloud more strongly:



2.8 Electron Affinity and Exceptions

Electron affinity generally becomes more negative across a period, but exceptions occur due to stable half-filled or filled configurations and small size.

Exception	Observation	Reason
N less negative than C/O	N resists electron gain	Stable half-filled $2p^3$ configuration
F less negative than Cl	Cl has more negative electron affinity	Very small F atom has high electron-electron repulsion
Noble gases	Positive or very low tendency	Stable closed shell

2.9 Electronegativity Scales

Electronegativity is not directly measured; it is assigned using scales. The Pauling scale is most common in basic chemistry.

Scale	Basis	Use
Pauling	Bond energy differences	Most common qualitative scale
Mulliken	Average of ionization energy and electron affinity	Connects to atomic properties
Allred-Rochow	Electrostatic attraction by nucleus	Uses effective nuclear charge and radius

2.10 Polarization and Fajan's Rules

Fajan's rules explain covalent character in ionic compounds. A small highly charged cation polarizes a large anion strongly, increasing covalent character.

Fajan's Rules

- Smaller cation has greater polarizing power.
- Higher cation charge has greater polarizing power.
- Larger anion is more easily polarized.
- Cations with pseudo-noble gas configuration polarize more strongly.

Solved Example: Covalent Character

AlCl_3 has more covalent character than NaCl because Al^{3+} is smaller and more highly charged than Na^+ , so it polarizes Cl^- more strongly.

2.11 Coordination Number and Geometry in Complexes

Coordination number is the number of ligand donor atoms directly attached to the metal ion. Geometry depends on coordination number and electronic arrangement.

Coordination number	Geometry	Typical comment
2	Linear	Common for Ag^+ and Cu^+ complexes
4	Tetrahedral	Common when ligands are bulky or weak-field
4	Square planar	Common for d^8 metal ions like Pt^{2+}
6	Octahedral	Most common geometry for transition-metal complexes

Previous-Year Style Questions

1. Explain effective nuclear charge and shielding with periodic trends.
2. Discuss trends in atomic radius, ionization energy and electronegativity.
3. Explain penetration of s , p , d , f orbitals and its effect on energy.
4. State HSAB principle and predict stable acid-base combinations.
5. Explain VSEPR shapes and coordination geometries with examples.

2.12 Common Exceptions and Exam Traps

Periodic trends are general rules, but exam questions often test exceptions. Mentioning the reason behind exceptions improves answer quality.

Trend exception	Observation	Reason
Be and B ionization energy	$IE(\text{Be}) > IE(\text{B})$	Electron removed from B enters higher-energy $2p$ orbital
N and O ionization energy	$IE(\text{N}) > IE(\text{O})$	N has stable half-filled $2p^3$ arrangement
Ga smaller than Al anomaly	Size does not increase as expected	Poor shielding by filled $3d$ electrons
Lanthanide contraction	Similar radii of $4d$ and $5d$ elements	Poor shielding by $4f$ electrons

Exam Answer: Why $IE(\text{N}) > IE(\text{O})$?

Nitrogen has electronic configuration $1s^2 2s^2 2p^3$, where the three p electrons are half-filled and relatively stable. Oxygen has $1s^2 2s^2 2p^4$, so one p orbital contains paired electrons. Removing one electron from oxygen reduces electron-electron repulsion, so oxygen has lower first ionization energy than nitrogen.

Quick Revision — Module 5

Periodic Properties | BS-CH101

Key Formulas at a Glance

Formula / Rule	Meaning
$Z_{\text{eff}} = Z - \sigma$	Effective nuclear charge felt by valence electron.
$s > p > d > f$	Penetration order within a shell.
$E(ns) < E(np) < E(nd) < E(nf)$	Orbital energy order for same principal shell in multi-electron atoms.
$IE_1 < IE_2 < IE_3$	Successive ionization energies increase.
$\mu = \alpha E$	Induced dipole moment relation; α is polarizability.

One-Line Definitions

- **Effective nuclear charge:** Net positive charge experienced by an electron.
- **Shielding:** Reduction of nuclear attraction due to inner electrons.
- **Penetration:** Ability of an orbital to approach the nucleus.
- **Ionization energy:** Energy required to remove an electron from gaseous atom.
- **Electron affinity:** Energy change when an electron is added to gaseous atom.
- **Electronegativity:** Tendency of atom to attract bonded electron pair.
- **HSAB principle:** Hard acids bind hard bases and soft acids bind soft bases.

Frequently Asked / Viva Questions

Q1. Define effective nuclear charge.

It is the net nuclear charge felt by an electron after shielding.

Q2. Why does atomic radius decrease across a period?

Z_{eff} increases and pulls electrons closer.

Q3. Why does ionization energy decrease down a group?

Atomic size and shielding increase, so valence electrons are removed more easily.

Q4. Write the penetration order.

$s > p > d > f$.

Q5. Why is an s orbital lower in energy than a p orbital of same shell?

The s orbital penetrates closer to the nucleus and feels greater Z_{eff} .

Q6. What is polarizability?

Frequently Asked / Viva Questions (contd.)

Ease with which an electron cloud is distorted by an electric field.

Q7. What is HSAB principle?

Hard acids prefer hard bases and soft acids prefer soft bases.

Q8. Give examples of hard and soft bases.

Hard bases: F^- , OH^- ; soft bases: I^- , S^{2-} .

Q9. What decides coordination geometry?

Coordination number, ligand field, metal ion size, and electron configuration.

Q10. What is VSEPR theory used for?

It predicts molecular geometry from repulsion among electron pairs.

Summary Reference Table

Trend	Reason	Result
Across period	Z_{eff} increases	Size decreases, IE increases
Down group	Shell number increases	Size increases, IE decreases
Penetration	s enters closest to nucleus	s orbital has lower energy
HSAB	Charge density and polarizability	Predicts stable acid-base pairs